MAPÚA MALAYAN DIGITAL COLLEGE

A college under Mapúa Malayan Colleges Laguna

Data Mining Principles Project Presentation

Presented by:

Trisha Mei Atienza

MAPÚA MALAYAN DIGITAL COLLEGE

A college under Mapúa Malayan Colleges Laguna

Data Mining Principles
Project Presentation

Review of Key Insights from Milestone 1

EDA Insights — Key Patterns Discovered

Milestone 1 · 993 customers across 3 datasets

Transaction amounts are highly right-skewed

Total spend ranges from ₱76 to ₱489,516 — a small premium tier drives most value. Outliers visible in boxplots.

Transaction types are compositional & differentiating

Bill payment, investment, loan payment, purchase sum to 1.0 per customer. Negatively correlated — high investors purchase less.

Customer engagement varies significantly

Active months: 1–7. Recency: 0–197 days. Strong signal of lifecycle stage and churn risk.

Satisfaction scores are stable but useful

Mean ~5.6 across all income groups. Likelihood-to-recommend slightly higher for purchase-oriented customers.

Income level reveals behavioral differences

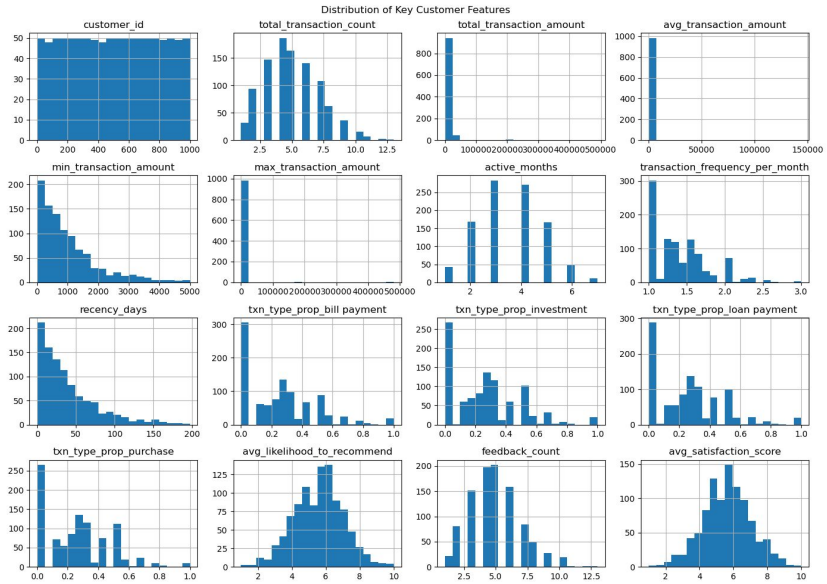
~250 Low / ~500 Medium / ~250 High. High-income customers show extreme transaction variability.

Surprising: zero missing values

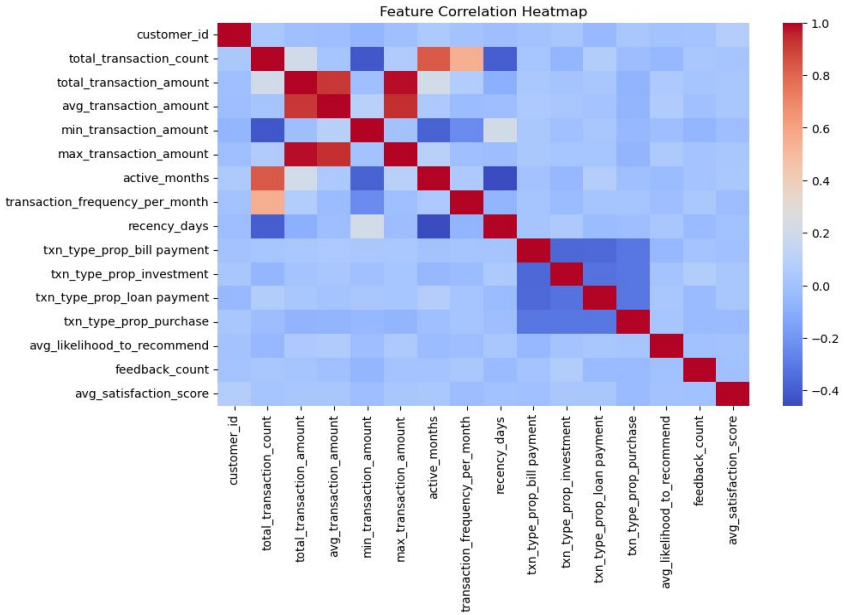
The engineered dataset (993 × 17) was fully complete — no imputation needed before analysis.

EDA Visuals

Feature distributions, correlations, and income-level breakdowns



Distribution of Key Customer Features



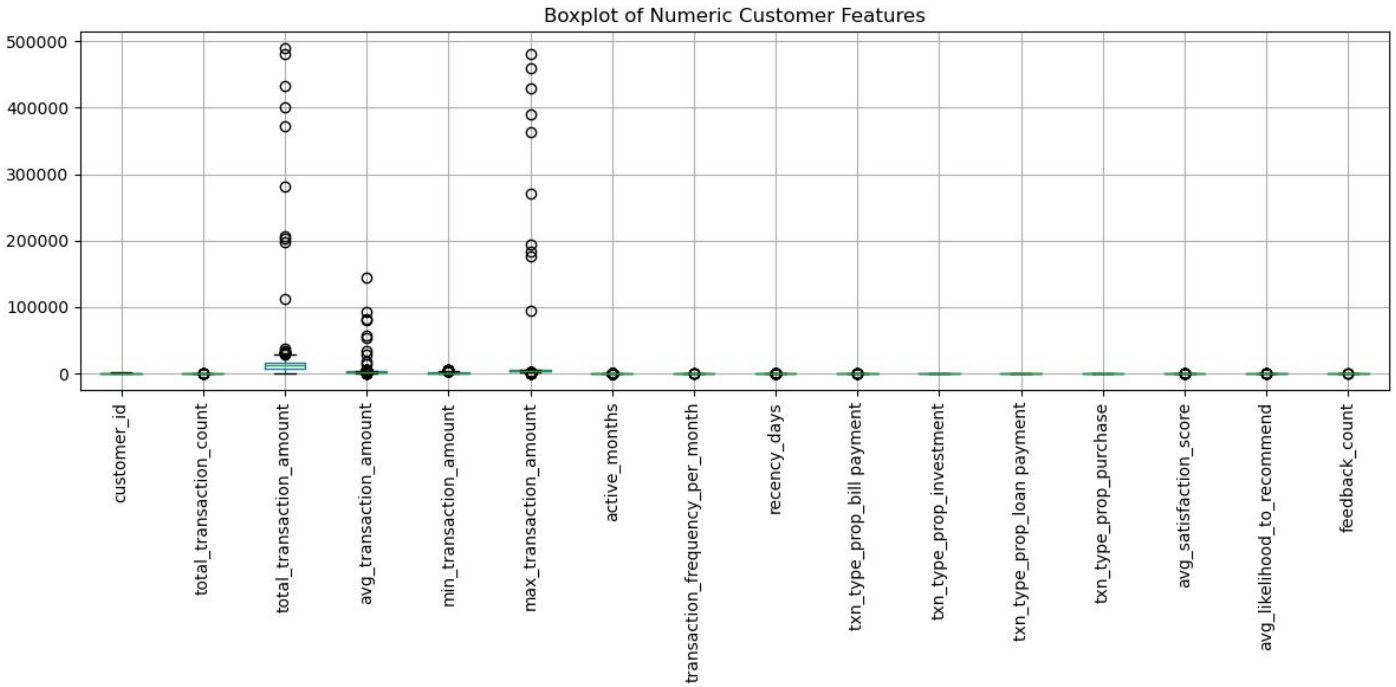
Distribution of Key Customer Features (histograms)

MAPUA MALAYAN DIGITAL COLLEGE

A college under Mapúa Malayan Colleges Laguna

EDA Visuals

Feature distributions, correlations, and income-level breakdowns



Boxplot of all features — Extreme outliers in transaction amounts

Feature Engineering — 11 Features Selected for Clustering

Guided directly by EDA findings

Transaction Value	Engagement	Behavioral (Txn Type)	Sentiment
total_transaction_amount	transaction_frequency_per_month	prop_bill_payment	avg_satisfaction_score
avg_transaction_amount	active_months	prop_investment	avg_likelihood_to_recommend
	recency_days	prop_loan_payment	
		prop_purchase	

Excluded: total_transaction_count (redundant), min/max amounts (outlier-dominated), last_transaction_date (replaced by recency_days), feedback_count, income_level (EDA only)

MAPÚA MALAYAN DIGITAL COLLEGE

A college under Mapúa Malayan Colleges Laguna

Data Mining Principles
Project Presentation

Clustering Methodology

Clustering Methodology — Algorithm Selection

K-Means with StandardScaler preprocessing

Why K-Means?

1 Structured numerical data

993 customers × 11 numerical features — ideal fit for distance-based clustering

2 Hard segment assignments

Each customer belongs to exactly one group — needed for business action

3 Interpretable centroids

Cluster means can be directly compared and labeled for business use

4 Efficient at this scale

Fast convergence on ~1,000 rows; n_init=10 ensures stable results

Why StandardScaler?

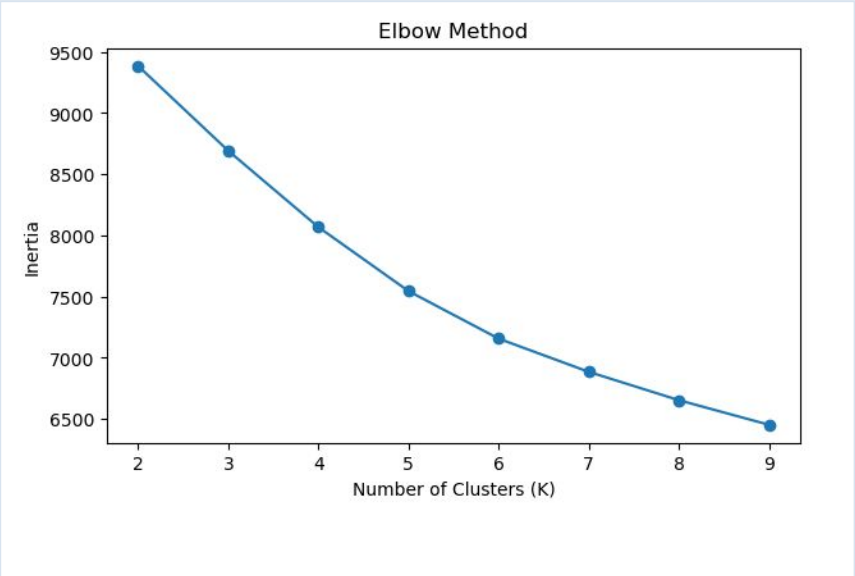
K-Means uses Euclidean distance. Without scaling, high-variance features dominate:

total_transaction_amount	1.13×10^9
recency_days	1,372
active_months	1.62
transaction_frequency	0.135

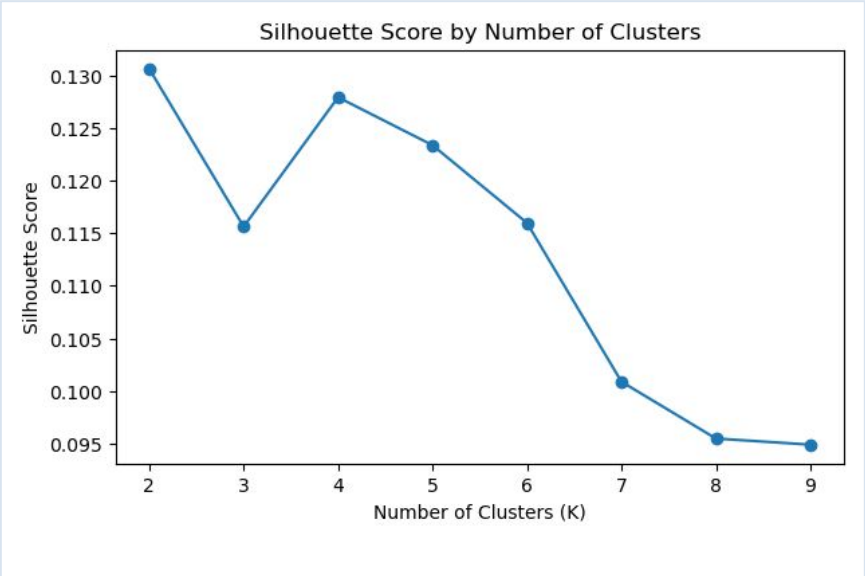
StandardScaler: zero mean, unit variance applied to all 11 features before fitting.

Parameter Tuning — Choosing K

Elbow method + silhouette score evaluated for K = 2 to 9



Elbow Method — Inertia by K



Silhouette Score by K

Why K = 3 was selected:

K=2 had the highest silhouette score (0.131) but collapses three distinct behavioral groups into two. K=3 provides interpretable segments (at-risk, core active, purchase-oriented) that are directly actionable for business strategy. The elbow plot also shows a clear inflection from K=2 to K=3.

MAPÚA MALAYAN DIGITAL COLLEGE

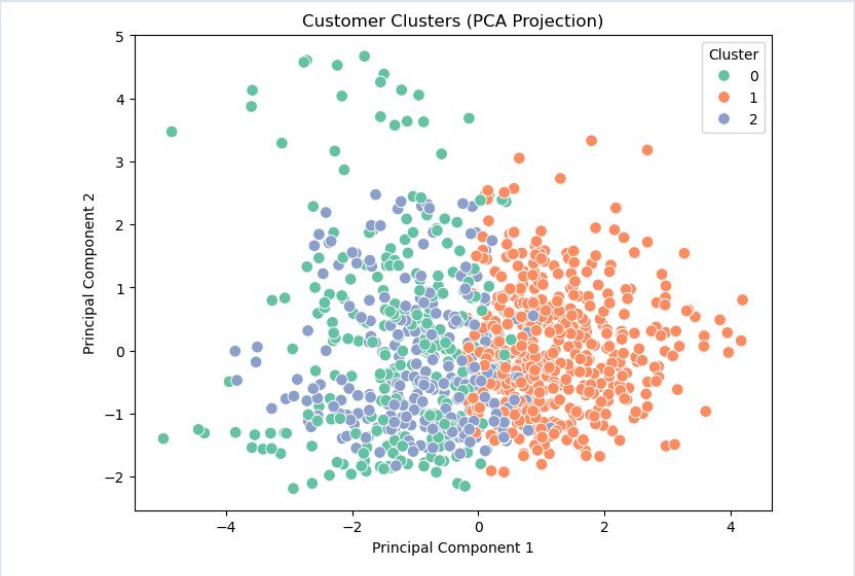
A college under Mapúa Malayan Colleges Laguna

Data Mining Principles
Project Presentation

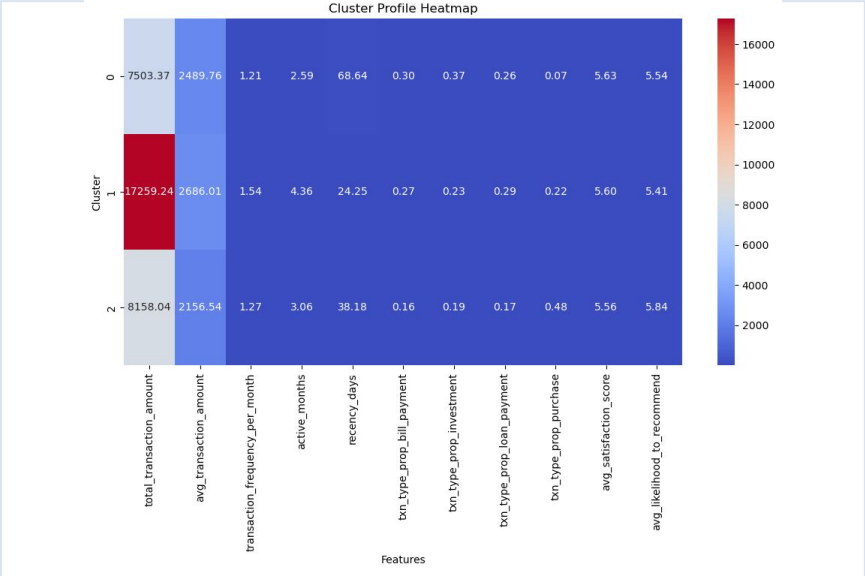
Cluster Analysis and Initial Interpretation

Cluster Analysis — Visualizing the Segments

PCA projection (2D) and cluster profile heatmap



PCA Projection — 3 clusters are spatially separable



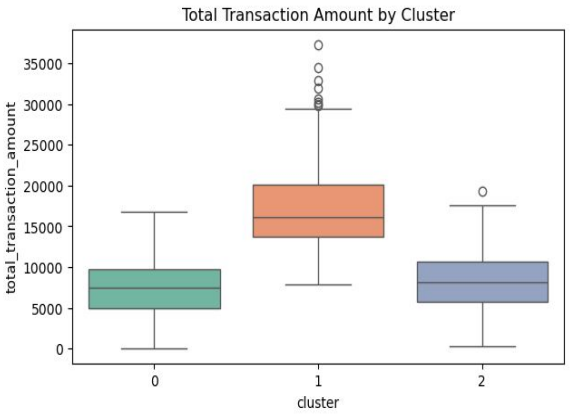
Cluster Profile Heatmap — Cluster 1 highest spend; Cluster 2 highest purchase share

MAPÚA MALAYAN DIGITAL COLLEGE

A college under Mapúa Malayan Colleges Laguna

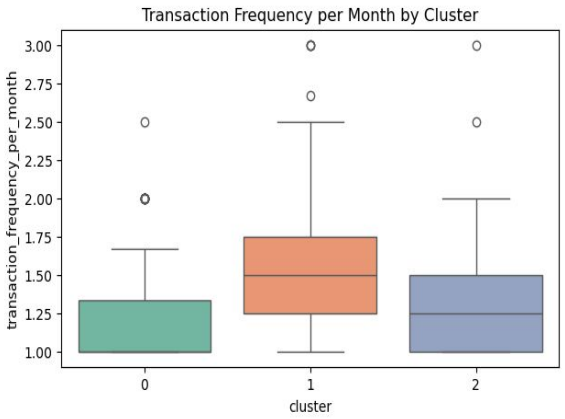
Cluster Comparison — Key Feature Distributions

Total spend, frequency, and recency across the 3 clusters



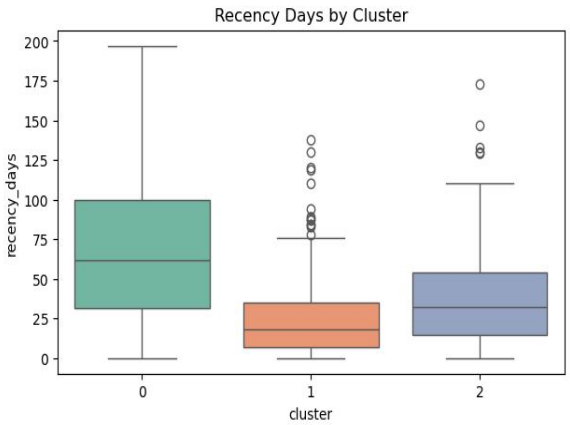
Total Transaction Amount by Cluster

Cluster 1 clearly highest — confirms high-value segment



Transaction Frequency per Month

Cluster 1 most active; Clusters 0 & 2 similar



Recency Days by Cluster

Cluster 0 highest recency (most disengaged, highest churn risk)

Key takeaway: Recency and total spend are the two sharpest cluster separators. Cluster 1 leads on both value and activity; Cluster 0 is the disengagement signal; Cluster 2 is distinguished by behavioral composition (purchase type) rather than volume.

MAPÚA MALAYAN DIGITAL COLLEGE

A college under Mapúa Malayan Colleges Laguna

Customer Segment Profiles

Three distinct, actionable customer groups identified

Cluster 0
Low-Value / At-Risk

251 customers (25.3%)

Avg Total Spend	₱7,503
Active Months	2.59
Recency	68.6 days
Top Txn Type	Investment (37%)

Lowest spend, fewest active months, highest recency. Strong churn signal. Past investment behavior suggests they were once engaged.

Re-engagement campaigns; personalized investment offers; win-back incentives.

Cluster 1
High-Value / Core Active

458 customers (46.1%)

Avg Total Spend	₱17,259
Active Months	4.36
Recency	24.3 days
Txn Frequency	1.54 / month

Highest spend, most active, lowest recency. FinMark's primary revenue drivers. Balanced transaction type mix.

Retention priority; upsell premium products; VIP service tiers.

Cluster 2
Purchase-Oriented Moderate

274 customers (27.6%)

Avg Total Spend	₱8,158
Purchase Share	48%
Recency	38.2 days
Likelihood Recommend	5.84 / 10

Moderate spend but dominant purchase behavior (48%). Highest NPS proxy score. Not visible through income-level EDA alone — discovered through clustering.

Product promotions; purchase bundling; referral programs leveraging high NPS.

MAPÚA MALAYAN DIGITAL COLLEGE

A college under Mapúa Malayan Colleges Laguna

Data Mining Principles
Project Presentation

Clustering Insights

Key Insights & Findings

What clustering revealed beyond the EDA

Value is concentrated in 46% of customers

Cluster 1 (458 customers) spends 2× more than the other groups — consistent with the EDA's right-skewed transaction distribution, now confirmed as a real, distinct segment.

Satisfaction doesn't differentiate clusters

Scores are nearly flat (5.56–5.63) across all groups. Sentiment is NOT a clustering driver, but Cluster 2's higher NPS (5.84) marks them as natural brand advocates.

Cluster 2 was invisible in EDA

Income-level analysis could not detect the purchase-oriented segment. Its defining trait — 48% purchase share — only emerged when all 11 features were jointly considered by K-Means.

Clusters are well-balanced (251/458/274)

No degenerate micro-clusters — all three segments are population-level groups suitable for targeted business campaigns.

Recency is the sharpest churn signal

Cluster 0's median recency (~65 days) vs Cluster 1's (~18 days) is the single clearest behavioral gap — stronger even than total spend differences.

MAPÚA MALAYAN DIGITAL COLLEGE

A college under Mapúa Malayan Colleges Laguna

Results at a Glance

993
Total Customers Analyzed

11
Features Selected for Clustering

K = 3
Final Cluster Count

0.116
Silhouette Score (K=3)

	Label	Size	Avg Total Spend	Key Trait	Action
0	Low-Value / At-Risk	251 (25%)	₱7,503	Highest recency (68.6d)	Re-engagement
1	High-Value / Core Active	458 (46%)	₱17,259	Highest spend & activity	Retention + upsell
2	Purchase-Oriented	274 (28%)	₱8,158	48% purchase share, high NPS	Promotions + referrals

Conclusion

What we achieved

A complete K-Means customer segmentation pipeline — data cleaning, EDA, feature selection, scaling, model fitting, and cluster interpretation.

Three clear segments

Low-Value At-Risk (25%), High-Value Core (46%), Purchase-Oriented Moderate (28%) — each with a distinct profile and recommended business action.

EDA informed clustering

A complete K-Means customer segmentation pipeline — data cleaning, EDA, feature selection, scaling, model fitting, and cluster interpretation.The 11 features selected directly from EDA patterns. Clustering confirmed EDA findings and revealed Cluster 2 — a behavioral group invisible in income-level analysis.